CS & IT ENGINEERING

Python For Data Science

Python Tokens

DPP 01 (Discussion Notes)



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[MCQ]

#Q. The output of below python code segment is _____.

```
a= 5.5
b = 2.0
c= 2
d = -2.0
e= a // b
f = d // c
print(e + f)
```

e = 5.5 // 2.0 ⇒ 2.75 ⇒ rounded off to lower bound Integer = 2

f = -2.0 // 2 ⇒ -1.0

= 2 + (-1.0) ⇒ 1.0

A 0

B 0.0

C 1.0 ✓

D -1.0

[MCQ]

(Lowest to highest)

#Q. The ascending order Of Precedence of below Operators is _____.

1. not in
2. <<
3. not
4. ^

not , not in , ^ (XOR) , << (Left shift)

3, 1, 4, 2

A 3, 2, 1, 4

B 3, 1, 2, 4

C ✓ 3, 1, 4, 2

D 2, 4, 1, 3

[NAT]

#Q. The result after evaluating the below expression in Python is 5

result =14 & 4 + 5 << 2 ^ 19 + 3 // 7 - 3 >> 4

9 36 16 0

// , + , - , << , >> , & , ^ , =

3//7 = 0

9+5 = 9

19+0 = 19

19-3 = 16

9 << 2 = 9 * 2^2

= 36

16 >> 4 ⇒ $16/2^4$ = 1

14 & 36 ⇒ 14 = 00001110

36 = 00010100

00000100 = 4

4 ^ 1 ⇒ 00000100

00000001

00000101 = 5

Result = 5

[MCQ]

#Q. Match The Following Operators with their associativity.

LIST-I	LIST-II
(A) ** (Exponentiation)	1. Left To Right
(B) & (Bitwise AND)	2. Right To Left
(C) is not (Identity)	
(D) = (Assignment)	

A-2, B-1, C-1, D-2

A. A-1, B-2, C-2, D-1

B. A-2, B-2, C-1, D-1

C. A-2, B-1, C-2, D-2

D. A-2, B-1, C-1, D-2

[NAT]

#Q. The output of below code segment is 210.

```
a = 0o63
b = a<<2
c = a>>3
print(b+c)
```

0o63 ⇒ Octal Integer

<< , >> operation Performed on Decimal numbers

204 + 6 = 210

1	0
6	3

$(63)_8 \Rightarrow 3\times8^0 + 6\times8^1$

$= 3 + 48$

$= 51$

$b = 51 << 2 = 51 * 2^2 = 51 * 4 = 204$

$c = 51 >> 3 = 51 // 2^3 = 51 // 8 = 6$

[MCQ]

#Q. What will be printed by below Python Code?

```
i = 0xAE1  # Hexa Decimal value
j = i & 152
k = j | 100
print(k)
```

j=128

k=228

& , | Performed on Decimal values

$(AE1)_{16}$ =

2	1	0
A	E	1

$1\times16^0 + E\times16^1 + A\times16^2$

$= 1 + 14*16 + 10\times256$

$= 1+224+2560$

$= 2785$

0000101011110000 1 = 2785

0000000010011000 = 152

0000000010000000 = 128

0000000001100100 = 100

0000000011100100 = 228

2	2785
2	1392 - 1
2	696 - 0
2	348 - 0
2	174 - 0
2	87 - 0
2	43 - 1
2	21 - 1
2	10 - 1
2	5 - 0
2	2 - 1
	1 - 0

A 344

B 0o344

C 0xe4

D ✓ 228

[MSQ]

ANS: A, D

#Q. What will be the value of result in the below code?

```
x = 0b(01010110)2 = 86
y = 0x123         = 291
z = 42            = 42
result = x + y - z
```

291 + 86 − 42

377 − 42 = $(335)_{10}$

A ✓ 517 in octal

B ✗ 517 in decimal

C ✗ 1f4 in hexa decimal

D ✓ 335 in in decimal

335 in octal

8	335	
8	41	−7
	5	−1

$= (517)_8$

2	1	0
1	2	3

$3\times16^0 + 2\times16^1 + 1\times16^2$

$= 3 + 32 + 256$

$= 291$

	291	
	(1)86	
	377	
	−42	
	335	

335 in hexa

16	335	
16	20	−F
	1	−4

$(14f)_{16}$

[MCQ]

#Q. What will be the value of k in the below code?

```
i = -13.5
j = 5
k = i % j
print(k)
```

-13.5 % 5

a % b when a, b signs are different == b - (a % b)

== 5 - (13.5 % 5)

== 5 - 3.5 = 1.5

A 0.0

B ✓ 1.5

C 3.5

D -1.5

THANK - YOU